2014 Final.

#1. $T = 24h = 86400\ s \Rightarrow \omega_0 = 2\pi/T = 7.2722e{-}05$ rad/sec

$(x_0, y_0, z_0) = (-110, 70, 10)$ km $\quad (\dot{x}_0, \dot{y}_0, \dot{z}_0) = (-10, 20, 1)$ m/sec

Two impulsive maneuver.

$t_f = 2h = 7200$ sec.

$(x_f, y_f, z_f) = 0$ km & $(\dot{x}_f, \dot{y}_f, \dot{z}_f) = 0$ m/sec

$\omega t_f = \pi/6 \Rightarrow \sin(\omega t_f) = 1/2$ & $\cos(\omega t_f) = \sqrt{3}/2$

$$\Rightarrow \dot{y}(0^+) = \frac{(6x_0(\omega t_f - \sin\omega t_f)) - y_0)\omega\sin(\omega t_f) - 2\omega x_0(4-3\cos(\omega t_f))(1-\cos\omega t_f)}{(4\sin(\omega t_f) - 3\omega t_f)\sin\omega t_f + 4(1-\cos(\omega t_f))^2}$$

$(6x_0(\omega t_f - \sin(\omega t_f)) - y_0)\omega\sin\omega t_f$

$= (6(-110)(\pi/6 - 1/2) - 70)\, 7.27722e{-}05 \cdot \frac{1}{2} \approx -3.1116e{-}03.$

$2\omega x_0(4-3\cos\omega t_f)(1-\cos\omega t_f)$

$= 2\times 7.27722e{-}05 \times (-110)(4 - 3\sqrt{3}/2)(1-\sqrt{3}/2) \approx -3.0049e{-}03.$

$\Rightarrow$ Numerator : $-1.0666e{-}04$

$(4\sin(\omega t_f) - 3\omega t_f)\sin\omega t_f + 4(1-\cos(\omega t_f))^2$

$= (4/2 - \pi/2)/2 + 4(1-\sqrt{3}/2)^2 \approx 0.2864$: Denominator.

$$\Rightarrow \dot{y}(0^+) = \frac{-1.0666e{-}04}{0.2864} \approx -3.7242e{-}04$$

$$\dot{x}(0^+) = -\frac{\omega x_0(4-3\cos\omega t_f) + 2(1-\cos\omega t_f)\dot{y}_0}{\sin\omega t_f}$$

$= -2\omega x_0(4 - 3\cos\omega t_f) - 4(1-\cos\omega t_f)\dot{y}_0$

$= +2\cdot 110 \cdot 7.27722e{-}05(4 - 3\sqrt{3}/2) + 4(1-\sqrt{3}/2)(3.7242e{-}04)$

$\approx 0.02263.$

$\dot{z}(0^+) = -z_0\,\omega\cot(\omega t_f)$

$= -10(7.27722e{-}05)\cot(\pi/6) \approx -1.2596e{-}03.$

$\Rightarrow \vec{\Delta v}_0 = (\dot{x}(0^+) - \dot{x}_0)\vec{i} + (\dot{y}(0^+) - \dot{y}_0)\vec{j} + (\dot{z}(0^+) - \dot{z}_0)\vec{k}$

$= (0.03263\vec{i} - 0.0204\vec{j} - 2.2596e{-}03\vec{k})$ km/s

$\Rightarrow \Delta v_0 = 0.03853$ km/s.

$\dot{x}(t) = \dot{x}_0\cos(\omega t) + (3x_0\omega + 2\dot{y}_0)\sin(\omega t)$

$\dot{x}(t_f) = 0.02263 \times \sqrt{3}/2 + (-330 \cdot 7.27722e{-}05 + 2\times(-3.7242e{-}04))/2.$

$\approx 7.2266e{-}03$ km/s $= 7.2266$ m/s

$\dot{y}(t) = (6x_0\omega + 4\dot{y}_0)\cos(\omega t) - 2\dot{x}_0\sin\omega t - 6\omega x_0 - 3\dot{y}_0$

$\dot{y}(t_f) = (-660 \cdot 7.27722e{-}05 + 4\times(-3.7242e{-}04))\sqrt{3}/2$

$\quad - 2\cdot 0.02263/2 + 660\times 7.27722e{-}05 + 3\times 3.7242e{-}04$

≈ -0.016377 km/s $= -16.3725$ m/s.

$\dot{z}(t) = \dot{z}_0\cos(\omega t) - \omega z_0\sin(\omega t)$

$\dot{z}(t_f) = -1.2596e{-}03 \times \sqrt{3}/2 - 7.27722e{-}05 \cdot 10/2 \approx -1.4545e{-}03$ km/s

$\Rightarrow -\vec{\Delta v}_f = (\dot{x}(t_f) - \dot{x}_f)\vec{i} + (\dot{y}(t_f) - \dot{y}_f)\vec{j} + (\dot{z}(t_f) - \dot{z}_f)\vec{k}$

$= (7.2266e{-}03\vec{i} - 0.016377\vec{j} - 1.4545e{-}03)$ km/s $\Rightarrow \Delta v_f = 0.01795$ km/s.

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#2. $\begin{cases} I_x\dot{\omega}_x = (I_y - I_z)\omega_y\omega_z + M_x \\ I_y\dot{\omega}_y = (I_z - I_x)\omega_x\omega_z + M_y \\ I_z\dot{\omega}_z = (I_x - I_y)\omega_x\omega_y + M_z \end{cases}$

A. $\omega_x = 0$. $\omega_y = 0$. $\omega_z = \bar{\omega}_z = const \Rightarrow \dot{\omega}_z = 0$. $\forall t \geq 0$

$$\Rightarrow \begin{cases} I_x\dot{\omega}_x = 0 \\ I_y\dot{\omega}_y = 0 \end{cases} \Rightarrow \omega_x \text{ and } \omega_y \text{ are constant.}$$

$$\Rightarrow \omega_x = \omega_y = 0. \quad \forall t \geq 0$$

$$\Rightarrow I_z\dot{\omega}_z = 0.$$

Since $\dot{\omega}_x = \dot{\omega}_y = \dot{\omega}_z = 0$, $\omega_x = 0 = \omega_y$, $\omega_z = \bar{\omega}_z$ is an equilibrium.

B. Let $\omega_x = 0 + \Delta\omega_x$ $\omega_y = 0 + \Delta\omega_y$ $\omega_z = \bar{\omega}_z + \Delta\omega_z$

$$\Rightarrow \begin{cases} I_x(\dot{0} + \Delta\dot{\omega}_x) = (I_y - I_z)(0 + \Delta\omega_y)(\bar{\omega}_z + \Delta\omega_z) + M_x \\ I_y(\dot{0} + \Delta\dot{\omega}_y) = (I_z - I_x)(0 + \Delta\omega_x)(\bar{\omega}_z + \Delta\omega_z) + M_y \\ I_z(\dot{\bar{\omega}}_z + \Delta\dot{\omega}_z) = (I_x - I_y)(0 + \Delta\omega_x)(0 + \Delta\omega_y) + M_z \end{cases}$$

$$\Rightarrow \begin{cases} I_x\Delta\dot{\omega}_x = (I_y - I_z)\Delta\omega_y\bar{\omega}_z + M_x \\ I_y\Delta\dot{\omega}_y = (I_z - I_x)\bar{\omega}_z\Delta\omega_x + M_y \\ I_z\Delta\dot{\omega}_z = 0. \end{cases}$$

C. $$\begin{bmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{bmatrix}\begin{bmatrix} \Delta\dot{\omega}_x \\ \Delta\dot{\omega}_y \\ \Delta\dot{\omega}_z \end{bmatrix} = \begin{bmatrix} 0 & (I_y - I_z)\bar{\omega}_z & 0 \\ (I_z - I_x)\bar{\omega}_z & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} \Delta\omega_x \\ \Delta\omega_y \\ \Delta\omega_z \end{bmatrix} + M.$$

$$\Rightarrow \begin{bmatrix} \Delta\dot{\omega}_x \\ \Delta\dot{\omega}_y \\ \Delta\dot{\omega}_z \end{bmatrix} = \begin{bmatrix} 0 & (I_y - I_z)\bar{\omega}_z/I_x & 0 \\ (I_z - I_x)\bar{\omega}_z/I_y & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} \Delta\omega_x \\ \Delta\omega_y \\ \Delta\omega_z \end{bmatrix} + M.$$

$$\Rightarrow \text{Characteristic equation : } s\left(s^2 - \frac{(I_y - I_z)(I_z - I_x)}{I_x I_y}\bar{\omega}_z^2\right) = 0.$$

$$\Rightarrow \text{Eigenvalues : } \lambda = 0 \text{ OR } \lambda = \pm\sqrt{\frac{(I_y - I_z)(I_z - I_x)}{I_x I_y}}\,\bar{\omega}_z.$$

If $I_y > I_z > I_x$ OR $I_y < I_z < I_x$, then all eigenvalues are real numbers
There exist real positive eigenvalue, thus the system is unstable
If $I_y = I_z$ OR $I_z = I_x$, then all eigenvalues are zero.
Hence the system is unstable.
If $I_y > I_z$ & $I_z < I_x$ OR $I_y < I_z$ & $I_z > I_x$, then there exist 2 complex eigenvalues, then the system is marginary stable.
(In this case, I_y can be same as I_x).

#3. $I_y\Delta\ddot{\theta} = 3\omega_0^2(I_z - I_x)\Delta\theta + M_y$.

A. $100\Delta\ddot{\theta} = 3\omega_0^2(200 - 150)\Delta\theta \Rightarrow \Delta\ddot{\theta} = 1.5\omega_0^2\Delta\theta$.

$$\Rightarrow \begin{bmatrix} \Delta\dot{\theta} \\ \Delta\ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1.5\omega_0^2 & 0 \end{bmatrix}\begin{bmatrix} \Delta\theta \\ \Delta\dot{\theta} \end{bmatrix} \Rightarrow \text{eigenvalues } \lambda = \pm\sqrt{\frac{3}{2}}\,\omega_0$$

Since there exist real positive eigenvalue, then the system is unstable whatever initial state is.

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B. Let $M_y = -k_p \Delta\theta - k_d \Delta\dot{\theta}$

$\Rightarrow I_y \Delta\ddot{\theta} = 3\omega_0^2(-I_x + I_z)\Delta\theta + M_y$

$\rightarrow 100\,\Delta\ddot{\theta} = 150\,\omega_0^2 \Delta\theta - k_p\Delta\theta - k_d\Delta\dot{\theta}$

$\Rightarrow \Delta\ddot{\theta} + 0.01\,k_d\,\Delta\dot{\theta} + (0.01\,k_p - 1.5\,\omega_0^2)\Delta\theta = 0.$

$\Rightarrow \{s^2 + 0.01 k_d s + (0.01\,k_p - 1.5\,\omega_0^2)\}\,\Delta\theta_s = 0$

$\Rightarrow 0.01 k_d = 0.0112$ & $0.01\,k_p - 1.5\omega_0^2 = 0.000064$

$\therefore k_d = 1.12$ & $k_p = 6.616e\text{-}03.$

$\therefore M_y = -6.616e\text{-}03\,\Delta\theta - 1.12\,\Delta\dot{\theta}.$

#4 Let $\mathfrak{z}^T = [\mathfrak{z}_1, \mathfrak{z}_2, \mathfrak{z}_3]$

$\Rightarrow \mathfrak{z}^T\mathfrak{z} + \mathfrak{z}_4^2$

$\frac{d}{dt}(\mathfrak{z}^T\mathfrak{z} + \mathfrak{z}_4^2) = 2\mathfrak{z}^T\dot{\mathfrak{z}} + 2\dot{\mathfrak{z}}_4\mathfrak{z}_4 = \mathfrak{z}^T S(\omega)\mathfrak{z} + \cancel{\mathfrak{z}_4\mathfrak{z}^T\omega} - \cancel{\mathfrak{z}_4\omega^T\mathfrak{z}}$

$= \mathfrak{z}^T S(\omega)\mathfrak{z} = \mathfrak{z}\cdot(\omega\times\mathfrak{z}) = \det([\mathfrak{z}, \omega, \mathfrak{z}]) = 0.$

$(\because)\ (a\times b)\cdot c = \det([a, b, c])$

Hence $\mathfrak{z}^T\mathfrak{z} + \mathfrak{z}_4^2$ is conserved.

Since $\mathfrak{z}^T\mathfrak{z} + \mathfrak{z}_4^2 = 1$ at $t=0$, then $\mathfrak{z}^T\mathfrak{z} + \mathfrak{z}_4^2 = 1 \ \forall\ t \geq 0$